

Agentic Robotics with ROS 2

Building Autonomous Task Execution from Natural Language Commands

Setup guide. Please complete this before the workshop.

The requirement, in one line

Ubuntu 22.04 with ROS 2 Humble. That is the whole requirement. Everything else in this guide is about how to get there from whatever you are running now.

We strongly recommend a Linux machine

If you have a computer that can run Ubuntu 22.04 directly, use it. Everything is faster, nothing is emulated, and graphics work without any extra setup. A spare laptop or a dual boot partition is worth the effort.

Windows and macOS both work through a Linux environment, and we give you one route for each below. They are well trodden, but they are more moving parts than running Linux directly.

What you need

- **8 GB of RAM.** 6 GB can be made to work, so tell us if that is you and we will sort something out.
- **15 GB of free disk.** ROS 2 comes to under a gigabyte, but we install more on the day, so leave the room.
- **An internet connection,** both for this guide and for the first half hour of the workshop, when we install the language models together. After that the system runs entirely offline.
- **No GPU.** None. Everything runs on the processor, by design. If you happen to have a graphics card it will simply be a little faster.
- **Basic Python, and a little ROS 2.** Nodes, topics, publishers. Everything beyond that is explained on the day.
- **No machine learning experience.** Genuinely none.

What to do before the workshop

Sections 1 to 4. Set aside about an hour, and do it a day or two early rather than on the morning. Most of that time is downloading, so you can leave it running.

Section 5 is a short check. Run it once you are finished and send us the output, so we know you are ready.

The language models are installed together at the start of the workshop, so there is nothing to do about those beforehand. Keep your internet connection available for the first half hour and we will walk through it line by line.

Where to start

Ubuntu 22.04 already	go straight to section 3
Windows	section 1, then 3 onward
macOS	section 2, then 3 onward
Ubuntu 24.04, or another Linux	email us first. 24.04 needs ROS 2 Jazzy, not Humble.

1. Windows

Recommended route: WSL2. It runs a real Ubuntu inside Windows, it is free, it is Microsoft's own supported feature, and graphical applications like RViz work without extra configuration.

You need Windows 10 version 2004 or later, or Windows 11.

1.1 Install Ubuntu 22.04

Open **PowerShell as Administrator** and run:

```
wsl --install -d Ubuntu-22.04
```

Reboot when asked. On first launch Ubuntu asks you to create a username and password. These are separate from your Windows login. Remember the password, you will need it for sudo.

1.2 Confirm it is WSL version 2

```
wsl --list --verbose
```

NAME	STATE	VERSION
* Ubuntu-22.04	Running	2

The VERSION column must read **2**. If it reads 1:

```
wsl --set-version Ubuntu-22.04 2
```

1.3 Turn on systemd

This lets background services start on their own, which we will need on the day. Inside your Ubuntu terminal:

```
sudo nano /etc/wsl.conf
```

Add these two lines, then save with Ctrl+O, Enter, Ctrl+X:

```
[boot]
systemd=true
```

Then back in PowerShell:

```
wsl --shutdown
```

Wait ten seconds and reopen Ubuntu.

From here on, work inside the Ubuntu terminal, not PowerShell. Continue at section 3.

Other options on Windows

A virtual machine running Ubuntu 22.04 in VirtualBox, VMware Workstation Player or Hyper-V works equally well. Dual booting Ubuntu is better still. WSL2 is written out in full simply because it is quickest to get working; any of these is fine and section 3 onward is identical.

If something goes wrong

wsl --install is not recognised

Windows is too old. Update it, or use a virtual machine.

Virtualisation errors

enable virtualisation in your BIOS, usually called VT-x, AMD-V or SVM.

RViz opens but is blank

update your graphics driver from the manufacturer's site, not Windows Update.

2. macOS

Recommended route: a virtual machine running Ubuntu 22.04. ROS 2 has no supported native macOS build, so a Linux virtual machine is the reliable path and gives you exactly the same environment as everyone else.

2.1 Install UTM

UTM is free and works on both Apple Silicon and Intel Macs. Download it from **mac.getutm.app**, or install with Homebrew:

```
brew install --cask utm
```

2.2 Get an Ubuntu 22.04 image

Apple Silicon	the Ubuntu 22.04 ARM64 entry in UTM's own gallery is the simplest option, and it is a guided install
Intel Mac	Ubuntu 22.04.5 LTS Desktop, the 64-bit PC image, from ubuntu.com/download/desktop

2.3 Create the machine

When UTM asks, choose **Virtualize**, not Emulate. Emulation is far too slow for this. Then allocate:

Memory	at least 8 GB
CPU cores	at least 4
Disk	at least 40 GB
Sharing	turn on clipboard sharing, it will save you a lot of typing

Install Ubuntu as normal and log in to the desktop.

From here on, work inside the Ubuntu virtual machine. Continue at section 3.

Other options on macOS

Parallels Desktop and VMware Fusion both work well if you already have one. A Docker container running ROS 2 also works, though getting RViz to display from a container on macOS takes extra setup with XQuartz. We have written out the UTM route because it is free and needs no display workarounds.

Expect it to run slower

A virtual machine shares your Mac's processor, so the language model will take longer to answer than it would natively. Ten to twenty-five seconds for an instruction is normal and nothing is broken. We will say so on the day so nobody worries.

If you cannot get a virtual machine working

Email us before the workshop. You will still be able to follow along, and you get all the code and data at the end to run afterwards.

3. ROS 2 Humble

Everyone does this section, inside Ubuntu 22.04. Paste each block in order. About 20 minutes, mostly downloading.

3.1 Language settings

```
sudo apt update && sudo apt install -y locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8
```

3.2 Add the ROS 2 software source

```
sudo apt install -y software-properties-common curl gnupg
sudo add-apt-repository universe -y

sudo curl -sSL \
  https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /usr/share/keyrings/ros-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(.
/etc/os-release && echo $UBUNTU_CODENAME) main" \
| sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

3.3 Install ROS 2 Humble

This is the long one. It installs to about 670 MB but downloads rather more, and takes a while.

```
sudo apt update
sudo apt install -y ros-humble-desktop
```

3.4 Install the extra packages the workshop uses

```
sudo apt install -y \
  ros-humble-ur-description \
  ros-humble-robot-state-publisher \
  ros-humble-joint-state-publisher-gui \
  ros-humble-xacro \
  ros-humble-tf2-tools \
  python3-colcon-common-extensions
```

ur_description is the Universal Robots UR5e model, the arm we will be controlling. The rest are the standard tools for describing a robot and drawing it.

3.5 Make ROS 2 available in every terminal

```
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
echo $ROS_DISTRO

humble
```

Note there is no `ros2 --version` command in Humble. It will give you an error. Use `echo $ROS_DISTRO` as above.

3.6 Check that ROS 2 really works

This confirms two ROS 2 programs can find and talk to each other, which is the foundation of everything on the day. In one terminal:

```
ros2 run demo_nodes_cpp talker
```

In a second terminal:

```
ros2 run demo_nodes_py listener
```

```
[talker]: Publishing: 'Hello World: 1'  
[listener]: I heard: [Hello World: 1]
```

Press Ctrl+C in both once you have seen it working.

If the listener stays silent, stop and email us. This needs sorting out beforehand, not on the day.

4. One Python package

```
pip install --user ikpy
```

That is the only one. ikpy works out the joint angles that put the robot's gripper where we want it. Everything else the workshop needs is already installed with ROS 2.

Check it, along with the two that came with ROS 2:

```
python3 -c "import rclpy, numpy, ikpy; print('all three imported')"  
all three imported
```

Do not use a Python virtual environment, and do not use Anaconda.

ROS 2 installs its Python parts into the system Python at /usr/bin/python3. A virtual environment cannot see them, and Anaconda takes priority over the system Python. Either one and nothing will work. Check with:

```
which python3  
/usr/bin/python3
```

If you get anything else, run `conda deactivate` or `deactivate`, or email us.

5. Check, and send us the result

Paste this whole block into your Ubuntu terminal and press Enter. It changes nothing, it only looks at what you have installed.

```
{
echo "--- system ---"
. /etc/os-release; echo "$PRETTY_NAME"; uname -m
echo "cores: $(nproc)"
free -h | awk 'NR==2{print "ram: "$2}'
df -h ~ | awk 'NR==2{print "free disk: "$4}'
grep -qi microsoft /proc/version && echo "running under WSL2"

echo; echo "--- ros 2 ---"
echo "ROS_DISTRO=${ROS_DISTRO:-NOT SOURCED}"
for pkg in ur_description robot_state_publisher \
          joint_state_publisher_gui rviz2 xacro tf2_ros; do
  ros2 pkg prefix $pkg >/dev/null 2>&1 \
    && echo "  OK      $pkg" || echo "  MISSING $pkg"
done

echo; echo "--- python ---"
python3 --version; which python3
echo "virtualenv: ${VIRTUAL_ENV:-none}"
for m in rclpy numpy ikpy; do
  python3 -c "import $m; print('  OK      $m')" 2>/dev/null \
    || echo "  MISSING $m"
done

} 2>&1
```

What a ready machine looks like

```
--- system ---
Ubuntu 22.04.5 LTS
x86_64
cores: 8
ram: 16Gi
free disk: 96G

--- ros 2 ---
ROS_DISTRO=humble
  OK      ur_description
  OK      robot_state_publisher
  OK      joint_state_publisher_gui
  OK      rviz2
  OK      xacro
  OK      tf2_ros

--- python ---
Python 3.10.12
/usr/bin/python3
virtualenv: none
  OK      rclpy
  OK      numpy
  OK      ikpy
```

Your core count, memory, disk and version numbers will differ. What matters is Ubuntu 22.04, ROS_DISTRO=humble, six OK lines under ros 2, python3 at /usr/bin/python3 with no virtualenv, and three OK lines under python.

Please send us this output at least two days before the workshop. If everything is OK, that is a one line email and we will know you are set. If something is missing, we can fix it with you beforehand rather than in front of everyone.

If something is missing

ROS_DISTRO=NOT SOURCED	you opened the terminal before doing section 3.5. Close it and open a new one.
a ROS package says MISSING	re-run the apt block in section 3.4.
python3 is not /usr/bin/python3	Anaconda or pyenv is taking priority. Run conda deactivate, or email us.
virtualenv is not none	run deactivate and try again.
ikpy MISSING	pip install --user ikpy
talker and listener do not see each other	email us. This one needs sorting before the workshop.

On the day

- Have your Ubuntu terminal open before we start.
- Keep your internet connection available. We install the language models together in the first half hour, about 3.7 GB.
- Two monitors help if you have them. One for the call, one for your terminal and the robot view.
- You get all the code and data at the end, to run again afterwards at your own pace.

Any questions, reply to the email this guide came with. We would much rather hear from you during the week than have you stuck on the morning of the workshop.